

100 Days of Linux - Day 061

Title: Practical Assignment - Environment Variables & Package Management

Today's Goal

Apply everything learned about environment variables, aliases, and package management in one realistic Linux administration workflow. No new commands are introduced today.

Why This Matters

Linux professionals configure environments, create productivity shortcuts, and install software every day. These skills are essential for developers, DevOps engineers, cloud engineers, and data engineers.

Commands Covered

env, printenv, echo, export, alias, unalias, apt

Scenario

You have joined a new project. Configure your terminal environment, create useful shortcuts, inspect system variables, and install the tools required for development.

Practical Assignment (5 Tasks)

1. Display all environment variables using **env**. Record the values of HOME, USER, PATH, and SHELL in a file named environment_report.txt.
2. Create three temporary environment variables named PROJECT, TEAM, and YEAR. Display them using **echo** and verify them with **printenv**.
3. Create aliases named **ll** (ls -lah), **cls** (clear), and **update** (sudo apt update && sudo apt upgrade). Test each alias, then remove the ll alias using **unalias**.
4. Use **apt search** to find the tree package. Install it (if needed), verify it works, then display package information using **apt show**.
5. Display your current working directory and list the files created during today's assignment.

Hints

Complete one task at a time. Verify each command before moving to the next step.

Mini Challenge

Configure your terminal for a new developer by creating two environment variables, three aliases, installing htop, and saving your aliases in ~/.bashrc. Reload the configuration using source ~/.bashrc.

Common Mistakes

Expecting exported variables or aliases to persist after closing the terminal without saving them in ~/.bashrc.

Did You Know?

Many development tools automatically read environment variables to find configuration values instead of hard-coding them.

Pro Tip

Keep a backup of your `~/.bashrc` before making major changes so you can easily restore your shell configuration.

Answer Key (Instructor Only)

Q1: `env && printenv HOME USER PATH SHELL > environment_report.txt` (or record manually)

Q2: `export PROJECT=Linux100 TEAM=Data YEAR=2026 && echo $PROJECT $TEAM $YEAR && printenv PROJECT TEAM YEAR`

Q3: `alias ll='ls -lah' && alias cls='clear' && alias update='sudo apt update && sudo apt upgrade' && ll && cls && unalias ll`

Q4: `apt search tree && sudo apt install tree && tree --version && apt show tree`

Q5: `pwd && ls`

Homework

Open a new terminal and recreate today's setup from memory. Then make your favorite aliases permanent in `~/.bashrc`.

Next Day Preview

Tomorrow you'll begin the Shell Scripting module by learning how to create, run, and understand your first Bash script.